

## CHITRAL, Pakistan

WMO: 415060

Lat: 35.883N

Lon: 71.800E

Elev: 4921

StdP: 12.26

Time Zone: 5.00 (E05)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 27.1       | 29.5 | 8.4                           | 10.1 | 47.5 | 12.2 | 12.3 | 46.6 | 20.2                  | 47.0 | 18.9 | 47.0 | 0.6                      | N/A  | 0.421 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 24.7                   | 99.3            | 69.3 | 97.1 | 69.1 | 94.9 | 68.5 | 75.1                | 91.8 | 73.2 | 90.1 | 71.7 | 88.7 | 6.5                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 70.5                            | 135.3 | 85.5 | 68.3 | 125.3 | 83.2 | 66.7 | 118.2 | 82.2 | 42.7          | 91.8 | 40.7 | 90.1 | 39.1 | 88.6 | 91.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 18.5              | 17.4 | 14.9 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |      |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |      |  |
|                   |      |      | WB                         | 21.3 | 79.0               | 3.2 | 3.6                                                | 19.0 | 81.6       | 17.2 | 83.7       | 15.4 | 85.7       | 13.1 | 88.3 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 61.1   | 40.2  | 42.7  | 51.5  | 61.0  | 69.3  | 78.4  | 83.0  | 79.7  | 72.0  | 60.9  | 49.7  | 43.5 |
|                                                                                  | DBStd       | 15.65  | 4.41  | 5.33  | 6.26  | 6.11  | 5.58  | 5.22  | 3.91  | 4.48  | 5.27  | 5.61  | 4.65  | 4.14 |
|                                                                                  | HDD50       | 844    | 305   | 212   | 58    | 3     | 0     | 0     | 0     | 0     | 0     | 60    | 206   |      |
|                                                                                  | HDD65       | 3273   | 770   | 624   | 420   | 148   | 21    | 1     | 0     | 1     | 7     | 155   | 459   | 667  |
|                                                                                  | CDD50       | 4894   | 0     | 8     | 104   | 332   | 600   | 853   | 1023  | 922   | 659   | 338   | 51    | 4    |
|                                                                                  | CDD65       | 1846   | 0     | 0     | 0     | 26    | 156   | 404   | 558   | 457   | 216   | 29    | 0     | 0    |
|                                                                                  | CDH74       | 23627  | 0     | 0     | 17    | 397   | 1945  | 5098  | 7448  | 5589  | 2621  | 504   | 8     | 0    |
|                                                                                  | CDH80       | 11983  | 0     | 0     | 0     | 103   | 808   | 2713  | 4240  | 2903  | 1067  | 149   | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.3    | 3.5   | 2.7   | 3.0   | 2.7   | 2.7   | 3.9   | 4.9   | 4.5   | 3.8   | 2.0   | 2.0   | 3.6  |
| Precipitation                                                                    | PrecAvg     | 21.2   | 1.9   | 3.3   | 4.4   | 3.6   | 1.9   | 0.6   | 0.5   | 0.5   | 0.7   | 1.1   | 1.4   | 1.3  |
|                                                                                  | PrecMax     | 29.8   | 5.7   | 8.9   | 10.2  | 6.3   | 5.0   | 1.8   | 3.6   | 1.7   | 4.3   | 9.4   | 7.6   | 4.7  |
|                                                                                  | PrecMin     | 9.7    | 0.0   | 0.4   | 0.7   | 0.6   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 4.1    | 1.6   | 2.1   | 2.4   | 1.4   | 1.3   | 0.5   | 0.7   | 0.4   | 0.9   | 1.8   | 1.5   | 1.4  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 58.7  | 66.3  | 76.0  | 88.0  | 94.4  | 102.1 | 102.9 | 100.2 | 94.0  | 89.5  | 74.8  | 63.1 |
|                                                                                  |             | MCWB   | 41.1  | 46.1  | 55.1  | 62.0  | 66.2  | 68.9  | 70.3  | 70.8  | 71.1  | 61.9  | 53.4  | 45.3 |
|                                                                                  | 2%          | DB     | 55.7  | 60.0  | 71.6  | 82.9  | 91.3  | 98.8  | 99.9  | 97.3  | 91.4  | 84.2  | 68.1  | 59.1 |
|                                                                                  |             | MCWB   | 39.9  | 44.7  | 52.3  | 59.9  | 64.6  | 66.9  | 70.5  | 71.0  | 69.1  | 61.1  | 51.2  | 43.7 |
|                                                                                  | 5%          | DB     | 52.9  | 56.8  | 68.0  | 78.5  | 88.5  | 95.9  | 97.5  | 94.9  | 89.1  | 79.8  | 64.8  | 56.1 |
|                                                                                  |             | MCWB   | 39.5  | 43.6  | 50.6  | 57.4  | 63.6  | 65.8  | 69.8  | 70.5  | 67.7  | 59.4  | 50.0  | 41.9 |
|                                                                                  | 10%         | DB     | 50.0  | 53.6  | 64.5  | 74.6  | 85.3  | 92.9  | 95.4  | 92.4  | 86.3  | 75.3  | 61.6  | 53.4 |
|                                                                                  |             | MCWB   | 38.3  | 42.0  | 48.7  | 56.0  | 62.2  | 64.9  | 69.2  | 69.8  | 66.4  | 57.5  | 48.5  | 40.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 44.2  | 49.4  | 56.8  | 64.0  | 71.1  | 72.7  | 79.2  | 77.2  | 76.6  | 68.2  | 56.7  | 48.2 |
|                                                                                  |             | MCDB   | 53.9  | 57.5  | 73.6  | 83.5  | 90.3  | 93.2  | 94.0  | 92.8  | 91.7  | 81.2  | 70.5  | 58.4 |
|                                                                                  | 2%          | WB     | 42.2  | 47.4  | 54.3  | 61.1  | 67.1  | 70.2  | 74.8  | 74.9  | 73.1  | 63.5  | 53.5  | 45.4 |
|                                                                                  |             | MCDB   | 51.4  | 56.1  | 67.7  | 79.2  | 87.2  | 90.8  | 93.9  | 90.6  | 87.9  | 79.1  | 63.8  | 55.2 |
|                                                                                  | 5%          | WB     | 40.6  | 45.5  | 52.2  | 59.2  | 64.8  | 68.3  | 72.9  | 73.3  | 69.9  | 60.5  | 51.6  | 43.5 |
|                                                                                  |             | MCDB   | 48.7  | 53.7  | 64.3  | 74.3  | 84.2  | 88.8  | 92.0  | 88.5  | 85.0  | 76.5  | 61.3  | 53.3 |
|                                                                                  | 10%         | WB     | 39.4  | 43.4  | 50.3  | 57.6  | 63.0  | 66.7  | 71.5  | 71.9  | 67.6  | 58.3  | 49.6  | 41.8 |
|                                                                                  |             | MCDB   | 47.3  | 51.0  | 61.4  | 70.9  | 81.4  | 88.1  | 89.5  | 87.8  | 82.8  | 73.0  | 59.0  | 51.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 17.5  | 16.8  | 19.8  | 23.0  | 26.5  | 27.4  | 24.7  | 24.5  | 27.1  | 27.2  | 22.6  | 19.5 |
|                                                                                  |             | MCDBR  | 23.0  | 22.2  | 26.0  | 30.4  | 32.1  | 31.0  | 27.8  | 26.9  | 29.4  | 33.4  | 28.7  | 24.5 |
|                                                                                  |             | MCWBR  | 11.2  | 10.2  | 10.7  | 12.3  | 12.7  | 9.7   | 8.1   | 7.6   | 10.6  | 12.8  | 14.2  | 12.0 |
|                                                                                  | 5% WB       | MCDBR  | 19.4  | 18.4  | 24.0  | 26.9  | 29.5  | 27.9  | 25.0  | 24.5  | 27.8  | 30.9  | 24.4  | 22.4 |
|                                                                                  |             | MCWBR  | 10.7  | 9.6   | 10.8  | 11.8  | 13.6  | 10.4  | 9.7   | 10.1  | 12.9  | 14.1  | 13.3  | 12.2 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.346  | 0.389 | 0.452 | 0.475 | 0.476 | 0.481 | 0.489 | 0.461 | 0.404 | 0.394 | 0.385 | 0.346 |      |
|                                                                                  | taud        | 1.998  | 1.867 | 1.773 | 1.834 | 1.917 | 1.939 | 1.949 | 2.046 | 2.169 | 2.093 | 2.013 | 2.059 |      |
|                                                                                  | Ebn at Noon | 259    | 260   | 255   | 258   | 259   | 257   | 254   | 259   | 269   | 258   | 242   | 250   |      |
|                                                                                  | Edh at Noon | 42     | 55    | 66    | 65    | 61    | 60    | 59    | 52    | 44    | 43    | 42    | 37    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 820    | 1053  | 1476  | 1833  | 2088  | 2301  | 2251  | 2082  | 1794  | 1360  | 926   | 762   |      |
|                                                                                  | RadStd      | 64     | 114   | 110   | 156   | 166   | 141   | 114   | 141   | 114   | 125   | 100   | 77    |      |

## Historical Trends

|      |                        | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page